

Windows Server Infrastructure Project

Client–Server Network Implementation for Cybertronic (Pvt) Ltd

Module	CNT 4008 — Computer Networks / Network Solution
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Project Snapshot

- 19+ endpoints across 5 departments, spanning Colombo and Kandy offices
- Windows Server 2012 R2 and Windows 7 Professional VMs deployed on VMware Workstation
- Active Directory Domain Services, DHCP, DNS, File Server, and Print Server implemented
- Domain: **ICBTLab.net** · Subnet: **192.168.56.0/24**

Executive Summary

This report documents the design and implementation of a client–server network infrastructure for Cybertronic, a software development company operating across Colombo and Kandy. The solution covers more than 19 endpoints across five departments and establishes centralised authentication, automated addressing, name resolution, and shared file/print resources on a Windows Server 2012 R2 platform, virtualised with VMware Workstation.

Key Deliverables

Deliverable	Status
Active Directory Domain (ICBTLab.net)	Configured
DHCP Server (192.168.56.0/24)	Deployed
DNS Forward / Reverse Lookup Zones	Configured
File Server with NTFS Permissions	Implemented
Print Server with Scheduled Access	Deployed

Deliverable	Status
Windows Firewall Security Rules	Configured

Technical Implementation

1. Virtual Machine Setup

Component	Specification
Hypervisor	VMware Workstation
Server OS	Windows Server 2012 R2
Client OS	Windows 7 Professional x64
Domain Name	ICBTLab.net
IP Range	192.168.56.0/24

Steps performed:

- Created two virtual machines — Windows Server 2012 R2 and Windows 7 Professional
- Configured host-only networking between the VMs
- Installed both operating systems from ISO images
- Created system snapshots at each stage as recovery points

2. Active Directory Domain Services

- Installed the AD DS server role via Server Manager
- Promoted the server to Domain Controller and created a new forest: **ICBTLab.net**
- Set the Administrator password for the domain controller: **123@com@LK**
- Created an Organizational Unit: **AllStaff**
- Created security groups: **ItAdmins**, **NonIt**
- Created user accounts: **user1** (ItAdmins), **user2** (NonIt)
- Enforced password change at next logon for new accounts

3. DHCP Server

Scope configuration

Parameter	Value
Scope Name	Icbthnddhcp
Start IP	192.168.56.1
End IP	192.168.56.254
Subnet Mask	255.255.255.0
Lease Duration	8 days

DHCP options configured

- 003 Router: 192.168.56.1
- 006 DNS Server: 192.168.56.1
- 015 DNS Domain Name: ICBTLab.net

Steps: Installed the DHCP Server role; created and activated the scope; configured server options; verified IP lease assignment on the client.

4. DNS Server

- Forward Lookup Zone created for **ICBTLab.net**
- Reverse Lookup Zone created for the **192.168.56.x** subnet
- PTR records created for reverse name resolution

Testing: Name resolution verified using nslookup; the DNS cache was cleared and the DNS service restarted whenever resolution issues occurred.

5. File Server

- Created a shared folder: C:\Shared, with sharing enabled and a user limit of 3
- Share comment set to "This is shared folder"
- Removed the default Read permission for Everyone
- Assigned NTFS permissions by group: **ItAdmins** — Full Control, **NonIt** — Read Only
- Disabled permission inheritance on the shared folder
- Verified client access via \\192.168.56.1\Shared

6. Print Server

- Added a virtual printer named **PrintServer** using the Microsoft PCL6 Class Driver
- Configured the printer port as File: (Print to File)
- Shared the printer under the name **PrintServer**
- Created a **PrinterGroup** in Active Directory and added authorised users
- Restricted access to a scheduled window: **9:00 AM – 6:00 PM**
- Assigned Print and Manage Documents permissions to the group

Network Security

Security Layer	Implementation
Authentication	Active Directory domain login
Firewall	Windows Firewall with a custom inbound ICMP rule
File Access	NTFS permissions (ItAdmins = Full Control, NonIt = Read Only)
Print Access	PrinterGroup restricted to scheduled hours (9 AM – 6 PM)
Disaster Recovery	VMware snapshots captured at each project milestone

Firewall Configuration

- Created a custom inbound rule for ICMPv4 traffic
- Rule action set to **Block the connection**
- Rule named **ICMPCustom**
- Verified enforcement by pinging the server from the client — connection successfully blocked

Monitoring & Troubleshooting

Tool	Purpose
ipconfig	View client IP configuration
ping	Test network connectivity
nslookup	DNS resolution testing
DHCP Console	View active address leases
services.msc	Manage Windows services

Common Issues Resolved

Issue	Resolution
DNS cache issues	Cleared cache and restarted the DNS service
Client IP not assigned	Restarted the DHCP service
Access denied	Added the user to the correct security group
Ping timeout	Modified the ICMP firewall rule

Conclusion

Successfully Delivered

- A virtualised server–client environment built with VMware Workstation
- Centralised authentication via Active Directory Domain Services
- Automated IP addressing through DHCP
- Resource sharing across File and Print servers
- Security enforcement via Windows Firewall and NTFS permissions
- Monitoring and troubleshooting workflows for common network issues

Skills Demonstrated

- Windows Server Administration
- Active Directory Management
- DHCP / DNS Configuration
- File & Print Server Deployment
- Network Security Implementation
- Troubleshooting & Technical Documentation

Appendices

Appendix A — System Configuration

Parameter	Value
Domain Name	ICBTLab.net
Server IP	192.168.56.1
DHCP Scope	192.168.56.1 – 192.168.56.254
Subnet Mask	255.255.255.0
DNS Server	192.168.56.1
Administrator Password	123@com@LK

Appendix B — User Accounts

Username	Group Membership
user1	ItAdmins, Domain Users
user2	NonIt, Domain Users

References

- Microsoft Corporation. (2013). *Windows Server 2012 R2: Active Directory Domain Services*. Microsoft TechNet.

- Microsoft Corporation. (2013). *DHCP Server Deployment Guide*. Microsoft TechNet.
- Microsoft Corporation. (2013). *File and Print Server Deployment*. Microsoft TechNet.
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